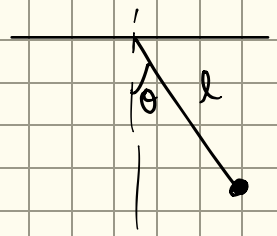

$x = l\sin\theta \Rightarrow \dot{x} = l\dot{\theta}\cos\theta$

$y = l\cos\theta \Rightarrow \dot{y} = -l\dot{\theta}\sin\theta$

$v^2 = \dot{x}^2 + \dot{y}^2 = l^2\dot{\theta}^2$

$$\mathcal{L} = \frac{1}{2}ml^2\dot{\theta}^2 + mgl\cos\theta$$

$$\frac{d}{dt}\frac{\partial\mathcal{L}}{\partial\dot{\theta}} = \frac{\partial\mathcal{L}}{\partial\theta} \Rightarrow \ddot{\theta} = -\frac{g}{l}\cos\theta$$

$L: TM \longrightarrow \mathbb{R}$ and we get equations of motion using Euler-lagrange equation. But these are second order differential equations.

Euler Lagrange: $\frac{d}{dt}\frac{\partial L}{\partial \dot{q}^i} = \frac{\partial L}{\partial q^i}$

Hamilton's Ideas: Classically $L = K - U$. Note K is homogenous of degree 2 ($K(tx) = t^2K(x)$) $2K = \dot{q}^i\frac{\partial K}{\partial \dot{q}^i} \Rightarrow 2dK = \frac{\partial K}{\partial \dot{q}^i}d\dot{q}^i + \dot{q}^i d\left(\frac{\partial K}{\partial \dot{q}^i}\right)$

$$dK = \frac{\partial K}{\partial q^i}dq^i + \frac{\partial K}{\partial \dot{q}^i}d\dot{q}^i$$

Subtracting $dK = \dot{q}^i d\left(\frac{\partial K}{\partial \dot{q}^i}\right) - \frac{\partial K}{\partial q^i}dq^i$ let $p_i = \frac{\partial K}{\partial \dot{q}^i}$

$$\Rightarrow dK = \dot{q}^i dp_i - \frac{\partial K}{\partial q^i}dq^i$$

Define $\tilde{K}$ to be a function of p_i and q^i given by rewriting $\dot{q}^i$ interms of p_i and subsing into K.

Note $d\tilde{K} = \frac{\partial \tilde{K}}{\partial q^i}dq^i + \frac{\partial \tilde{K}}{\partial p_i}dp_i$ firstly $\dot{q}^i = \frac{\partial \tilde{K}}{\partial p_i}$ and $\frac{\partial \tilde{K}}{\partial q^i} = -\frac{\partial K}{\partial q^i}$

Assume U is independent of p_i.

$$\dot{q}^i = \frac{\partial \tilde{K}}{\partial p_i} + \frac{\partial U}{\partial p_i}$$

$= \frac{\partial H}{\partial p_i}$ where $H = \tilde{K} + U$ $\leftarrow$ This is the Hamiltonian.

$$\dot{p}_i = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}^i}\right) = \frac{\partial L}{\partial q^i}$$

$$= \frac{\partial K}{\partial q^i} - \frac{\partial U}{\partial q^i}$$

$$= -\frac{\partial \tilde{K}}{\partial q^i} - \frac{\partial U}{\partial q^i}$$

$$= -\frac{\partial H}{\partial q^i}$$

$$\tilde{L} = \tilde{K} - U = \tilde{K} - H + \tilde{K} \Rightarrow \tilde{L} + H = 2\tilde{K} = \dot{q}^i \frac{\partial K}{\partial \dot{q}^i} = \dot{q}^i p_i$$

$\uparrow$ L but a function of p and q

$$\Rightarrow H = \dot{q}^i p_i - \tilde{L}$$ ← The legendre transform

Example: $L = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2$

$p = \frac{\partial L}{\partial \dot{x}} = m\dot{x}$ moreover $\dot{x} = \frac{p}{m} \Rightarrow \dot{x}^2 = \frac{p^2}{m^2}$

$$\tilde{L} = \frac{1}{2}m\frac{p^2}{m^2} + \frac{1}{2}kx^2 = \frac{p^2}{2m} + \frac{1}{2}kx^2$$

$$\dot{x} = \frac{\partial H}{\partial p} = \frac{p}{m}$$

$$\dot{p} = -\frac{\partial H}{\partial x} = -kx$$

Assume L has no explicit time dependence then

$$\frac{dL}{dt} = \frac{\partial L}{\partial q^i}\dot{q}^i + \frac{\partial L}{\partial \dot{q}^i}\ddot{q}^i$$

$$= \dot{q}^i \frac{d}{dt}\frac{\partial L}{\partial \dot{q}^i} + \ddot{q}^i \frac{\partial L}{\partial \dot{q}^i}$$

$$= \frac{d}{dt}\left(\dot{q}^i \frac{\partial L}{\partial \dot{q}^i}\right) \Rightarrow \frac{d}{dt}\underbrace{\left(\dot{q}^i p_i - L\right)}_{\text{Hamiltonian}} = 0$$

Example: $H = \frac{1}{2m}p^2 + \frac{1}{2}kx^2$ and $\dot{x} = \frac{p}{m}$, $\dot{p} = \frac{-k}{x}$

$$\frac{d}{dt}\begin{bmatrix} x \\ p \end{bmatrix} = \begin{bmatrix} 0 & 1/m \\ -k & 0 \end{bmatrix}\begin{bmatrix} x \\ p \end{bmatrix} \Rightarrow \begin{bmatrix} x \\ p \end{bmatrix} = \exp\left(t\begin{bmatrix} 0 & 1/m \\ -k & 0 \end{bmatrix}\right)\begin{bmatrix} x(0) \\ p(0) \end{bmatrix}$$

$$= \begin{bmatrix} \cos(\omega t) & -\frac{\omega}{k}\sin(\omega t) \\ m\omega\sin(\omega t) & \cos(\omega t) \end{bmatrix}\begin{bmatrix} x(0) \\ p(0) \end{bmatrix} \qquad \omega^2 = \frac{k}{m}$$

Lagrangian: $\ddot{x} = -\frac{kx}{m}$

Phase space will have coordinates q^i and p_i hence we take phase space to be T^*M

If I look at Hamiltonian H and vector field $X_H = \frac{\partial H}{\partial p_i}\frac{\partial}{\partial q^i} - \frac{\partial H}{\partial q^i}\frac{\partial}{\partial p_i}$

then flow solves Hamilton's equations of motion.

We get an (abelian) action of $C^\infty(T^*M)$ onto itself by $H \mapsto X_H$ since

$$X_H f = \frac{\partial H}{\partial p_i}\frac{\partial f}{\partial q^i} - \frac{\partial H}{\partial q^i}\frac{\partial f}{\partial p_i} =: \{H, f\}$$

$H \mapsto \{H, \cdot\}$ this is a derivation and antisymmetric $\Rightarrow$ biderivation.

Exercise: $\{f, \{g,h\}\} + \{g, \{h,f\}\} + \{h, \{f,g\}\} = 0$.

Def: let M be a smooth manifold. A poisson bracket $\{\cdot,\cdot\}$ on M is a map $C^\infty(M) \times C^\infty(M) \to C^\infty(M)$ s.t. (i) It linear over $\mathbb{R}$
(ii) Antisymmetric
(iii) Derivation in an entry
(iv) Satisfies Jacobi Identity.

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let $f \in C^\infty(T^*M)$ then f is conserved iff $X_H f = 0$.

Def: let $(M, \{\cdot,\cdot\})$ be a Poisson manifold if $\{f,g\} = 0$ $\forall g$ then f is said to be a casimir.

$$dH = \frac{\partial H}{\partial q^i} dq^i + \frac{\partial H}{\partial p_i} dp_i \qquad X_H = \frac{\partial H}{\partial p_i}\frac{\partial}{\partial q^i} - \frac{\partial H}{\partial q^i}\frac{\partial}{\partial p_i}$$

If we replace $dp_i \mapsto \frac{\partial}{\partial q^i}$ and $dq^i \mapsto -\frac{\partial}{\partial p_i} \Rightarrow dH \mapsto X_H$

$\omega = dp_i \wedge dq^i$ then $\iota_{X_H}\omega = -\frac{\partial H}{\partial p_i} dp_i - \frac{\partial H}{\partial q_i} dq^i = -dH$

$d\omega = 0$ moreover $\omega^n = n!\, dp_1 \wedge \cdots \wedge dp_n \wedge dq^1 \wedge \cdots \wedge dq^n$ and so ω is nondegenerate.

Exercise: $\omega(X_f, X_g) = \{f, g\}$. Use this to show $d\omega = 0$ leads to the Jacobi identity.

Def: Let M be a manifold. $\omega \in \Omega^2(M)$ is called symplectic if $d\omega = 0$ and ω_p is nondegenerate $\forall p \in M$.

Exercise: Show M must be even dimensional.

If ω is exact i.e. $\omega = -d\theta$ we call θ a symplectic potential.